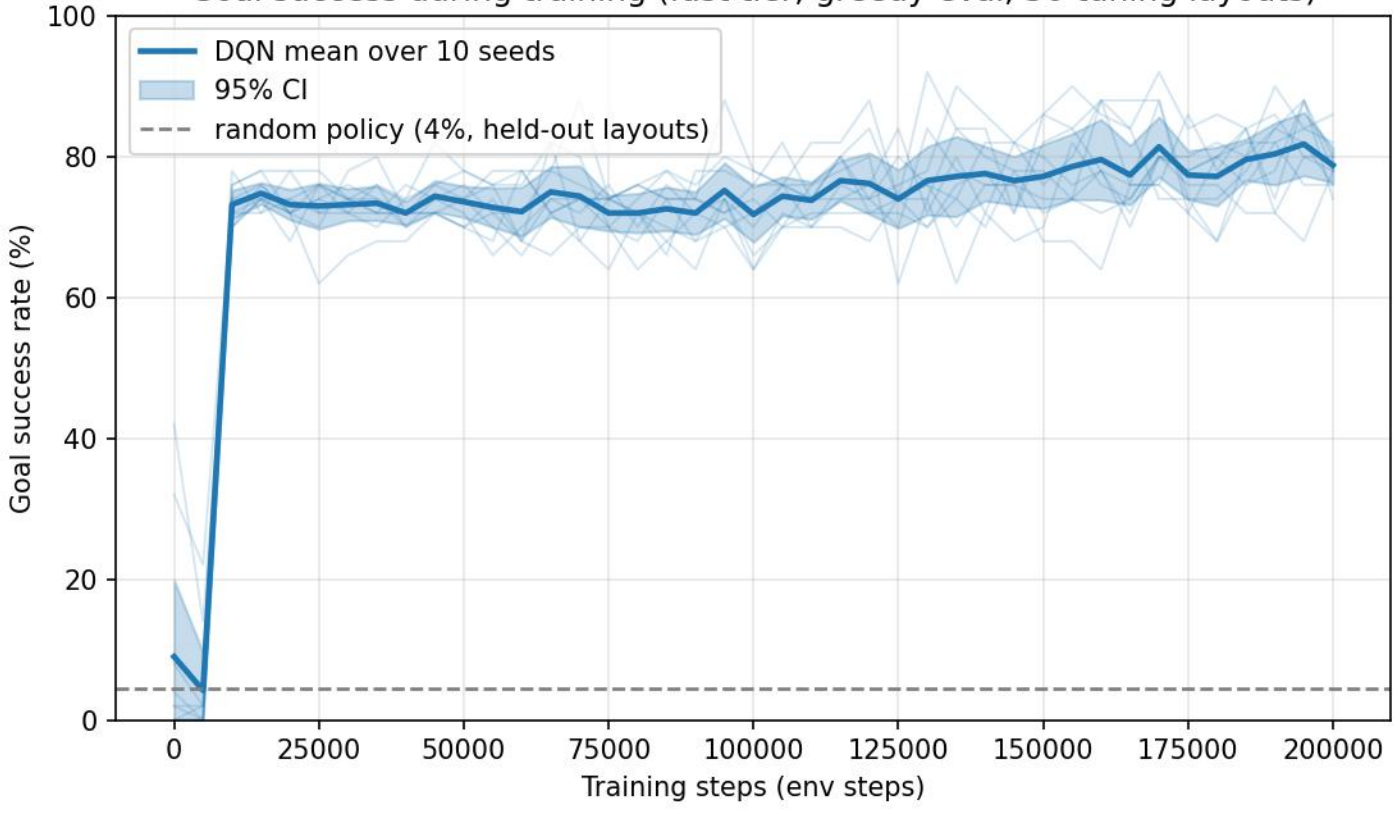
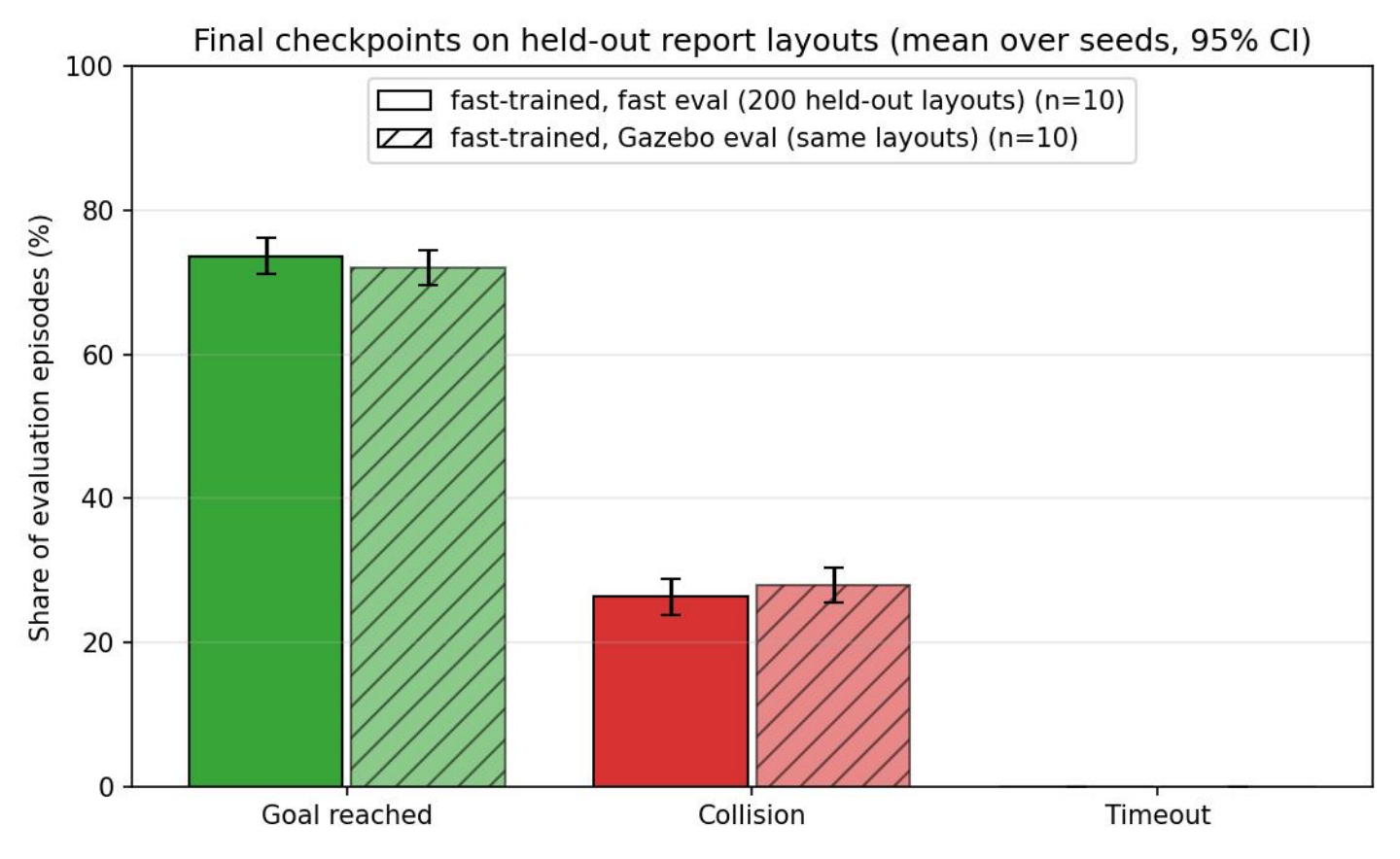
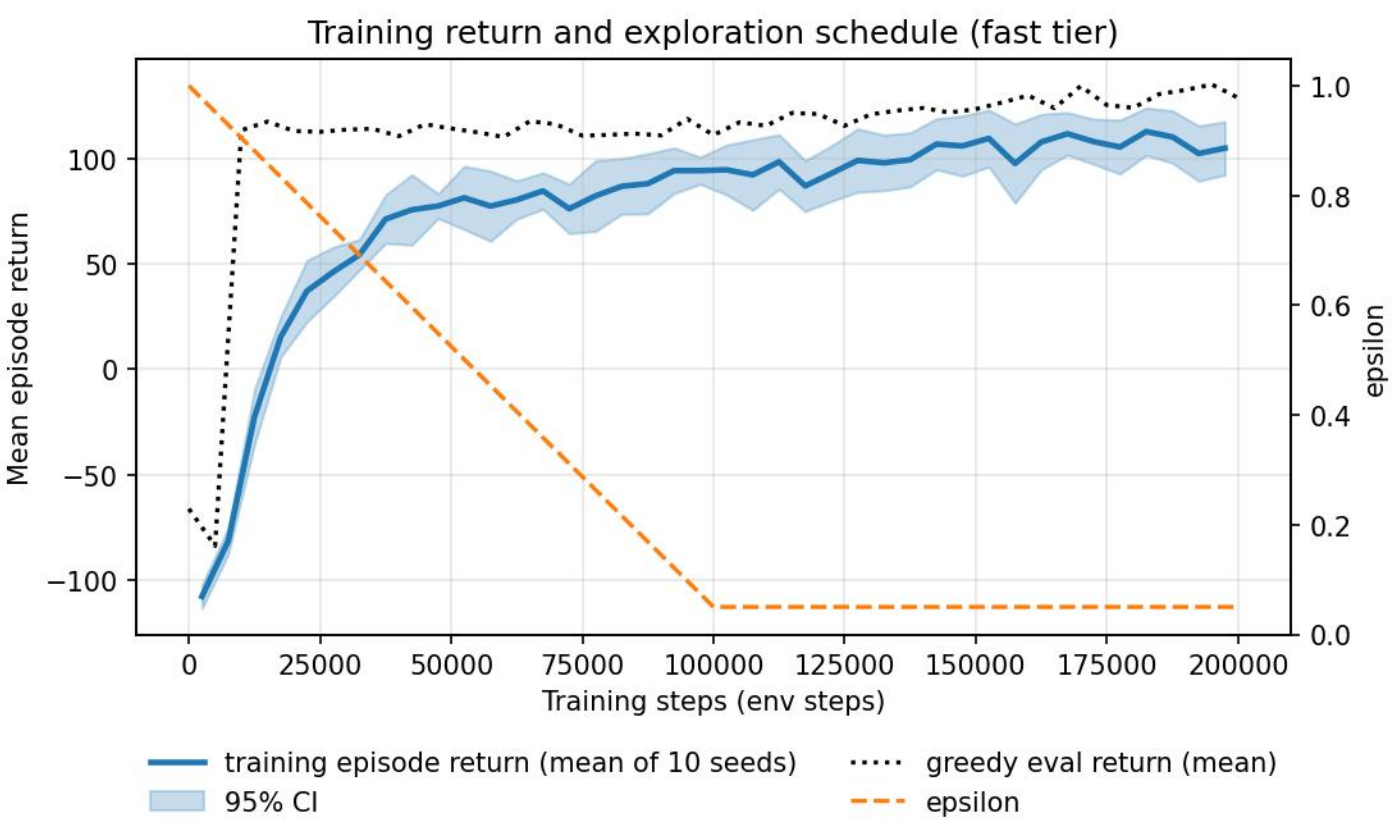
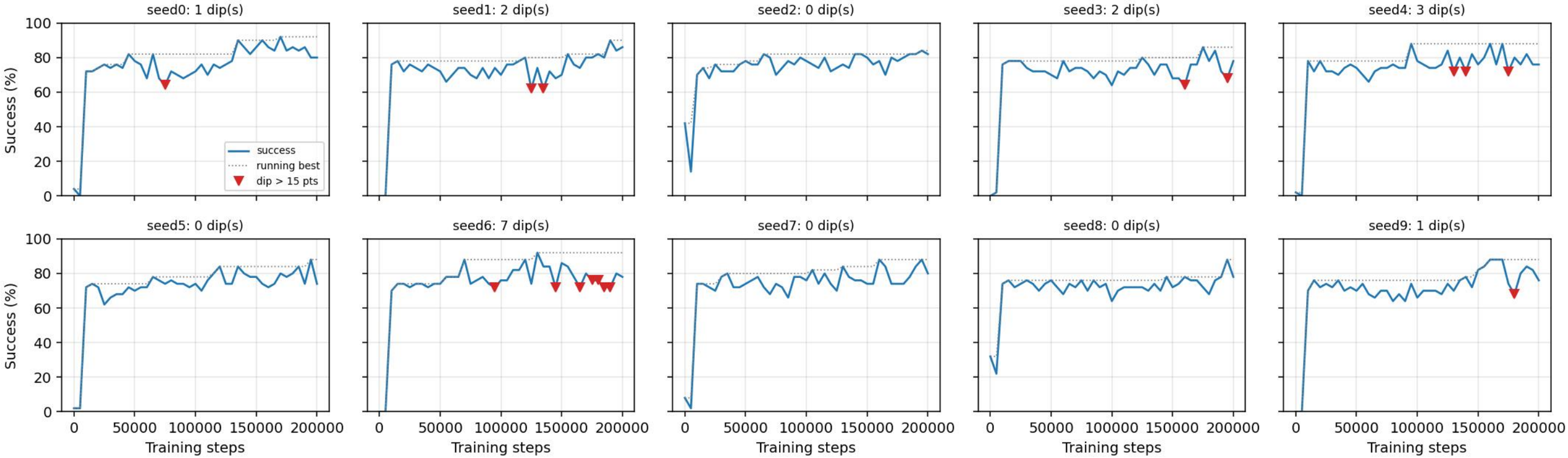
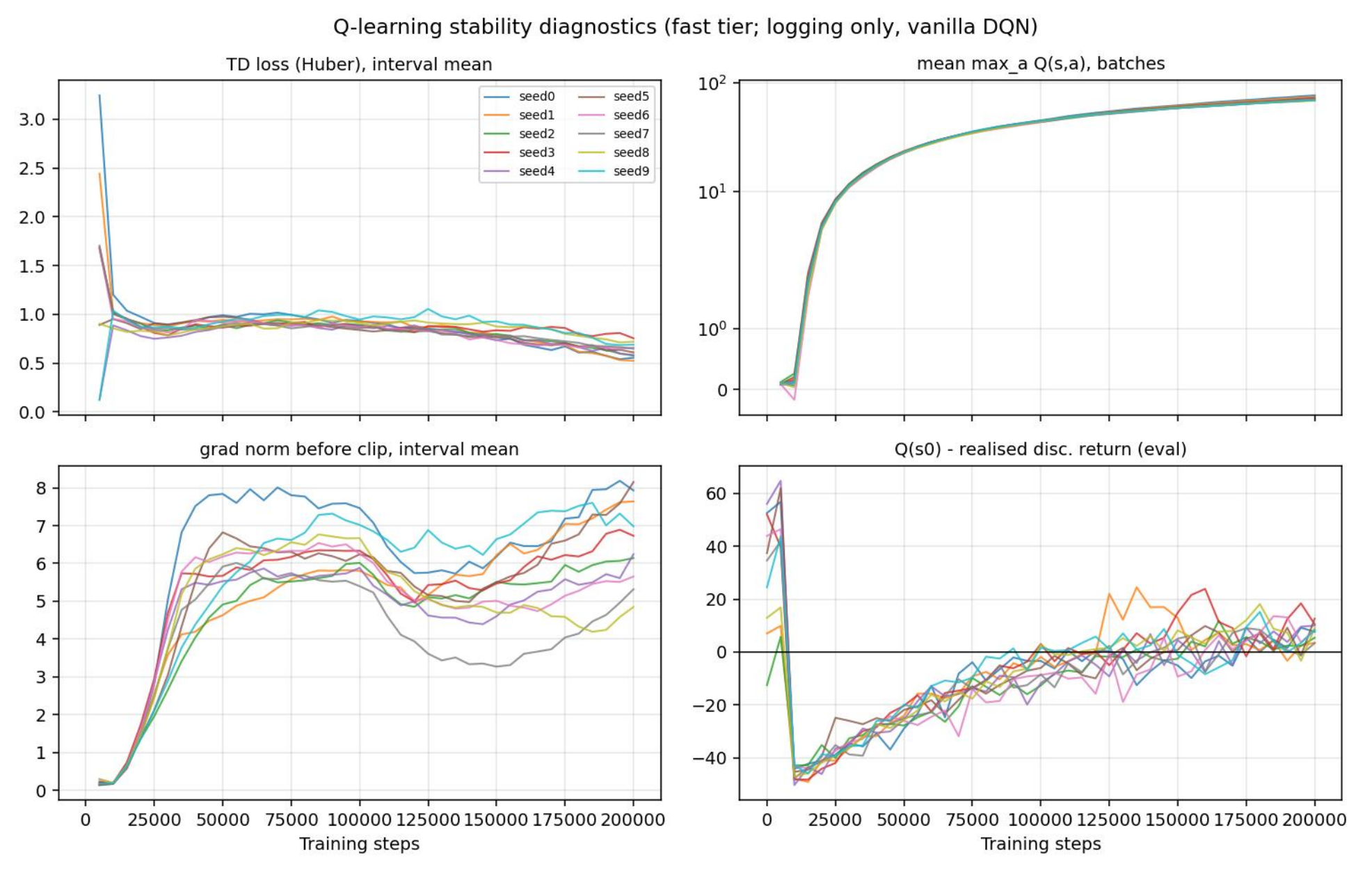
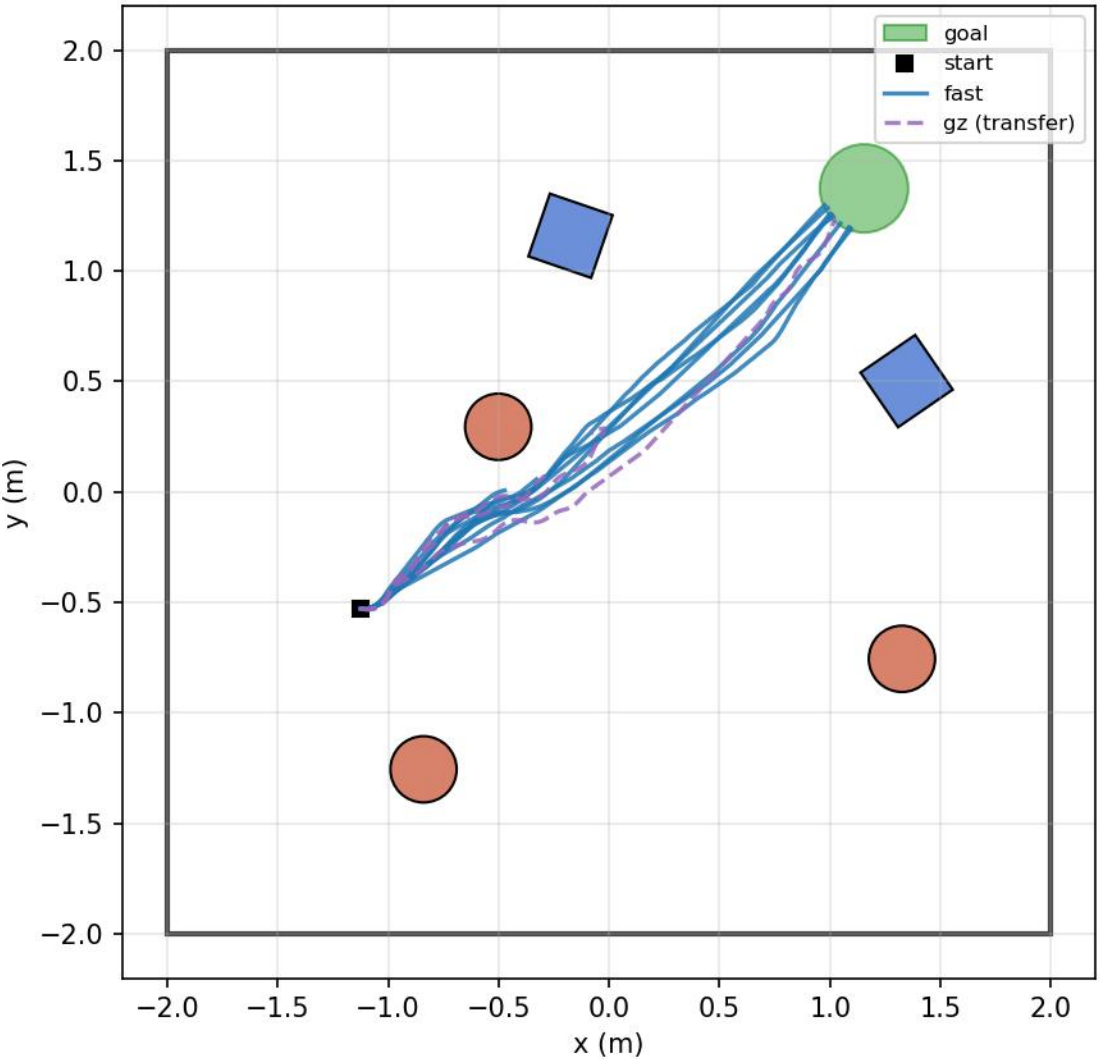

Goal success during training (fast tier, greedy eval, 50 tuning layouts)
DQN mean over 10 seeds
95% CI
random policy (4%, held-out layouts)
Goal success rate (%)
Training steps (env steps)
0
20
40
60
80
100
0
25000
50000
75000
100000
125000
150000
175000
200000


Final checkpoints on held-out report layouts (mean over seeds, 95% CI)
fast-trained, fast eval (200 held-out layouts) (n=10)
fast-trained, Gazebo eval (same layouts) (n=10)
Share of evaluation episodes (%)
100
80
60
40
20
0
Goal reached
Collision
Timeout


Training return and exploration schedule (fast tier)
Mean episode return
epsilon
Training steps (env steps)
100
50
0
−50
−100
1.0
0.8
0.6
0.4
0.2
0.0
0
25000
50000
75000
100000
125000
150000
175000
200000
training episode return (mean of 10 seeds)
95% CI
greedy eval return (mean)
epsilon


Per-seed goal success during training (fast tier, 50 tuning layouts per eval; dips counted after the 5k warm-up)
seed0: 1 dip(s)
seed1: 2 dip(s)
seed2: 0 dip(s)
seed3: 2 dip(s)
seed4: 3 dip(s)
seed5: 0 dip(s)
seed6: 7 dip(s)
seed7: 0 dip(s)
seed8: 0 dip(s)
seed9: 1 dip(s)
success
running best
dip > 15 pts
Success (%)
Training steps
0
20
40
60
80
100
0
50000
100000
150000
200000


Q-learning stability diagnostics (fast tier; logging only, vanilla DQN)
TD loss (Huber), interval mean
seed0
seed1
seed2
seed3
seed4
seed5
seed6
seed7
seed8
seed9
mean max_a Q(s,a), batches
grad norm before clip, interval mean
Q(s0) - realised disc. return (eval)
Training steps
Training steps


Trajectories on fixed layout 0 (one line per seed)
goal
start
fast
gz (transfer)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
y (m)
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
x (m)